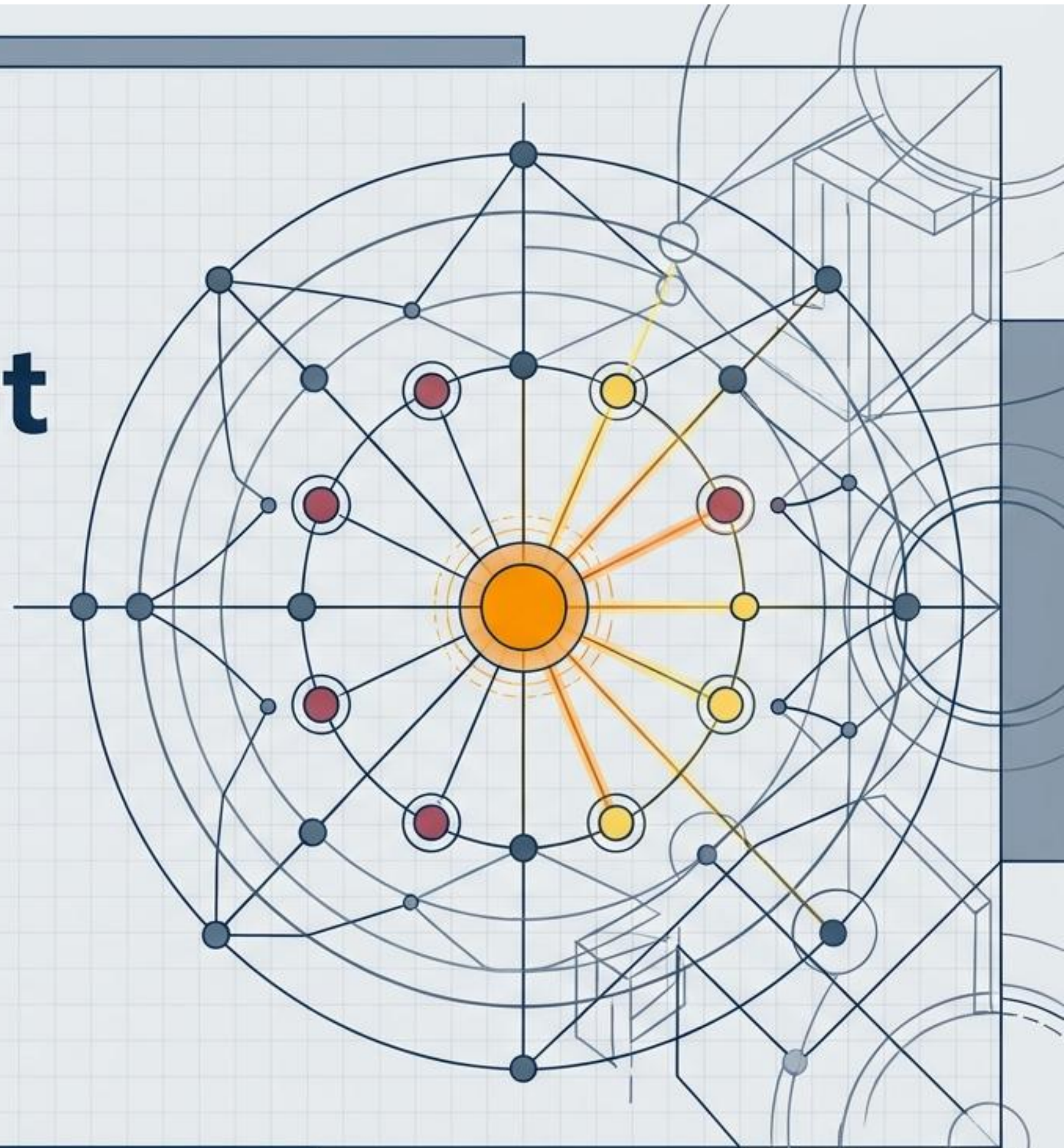
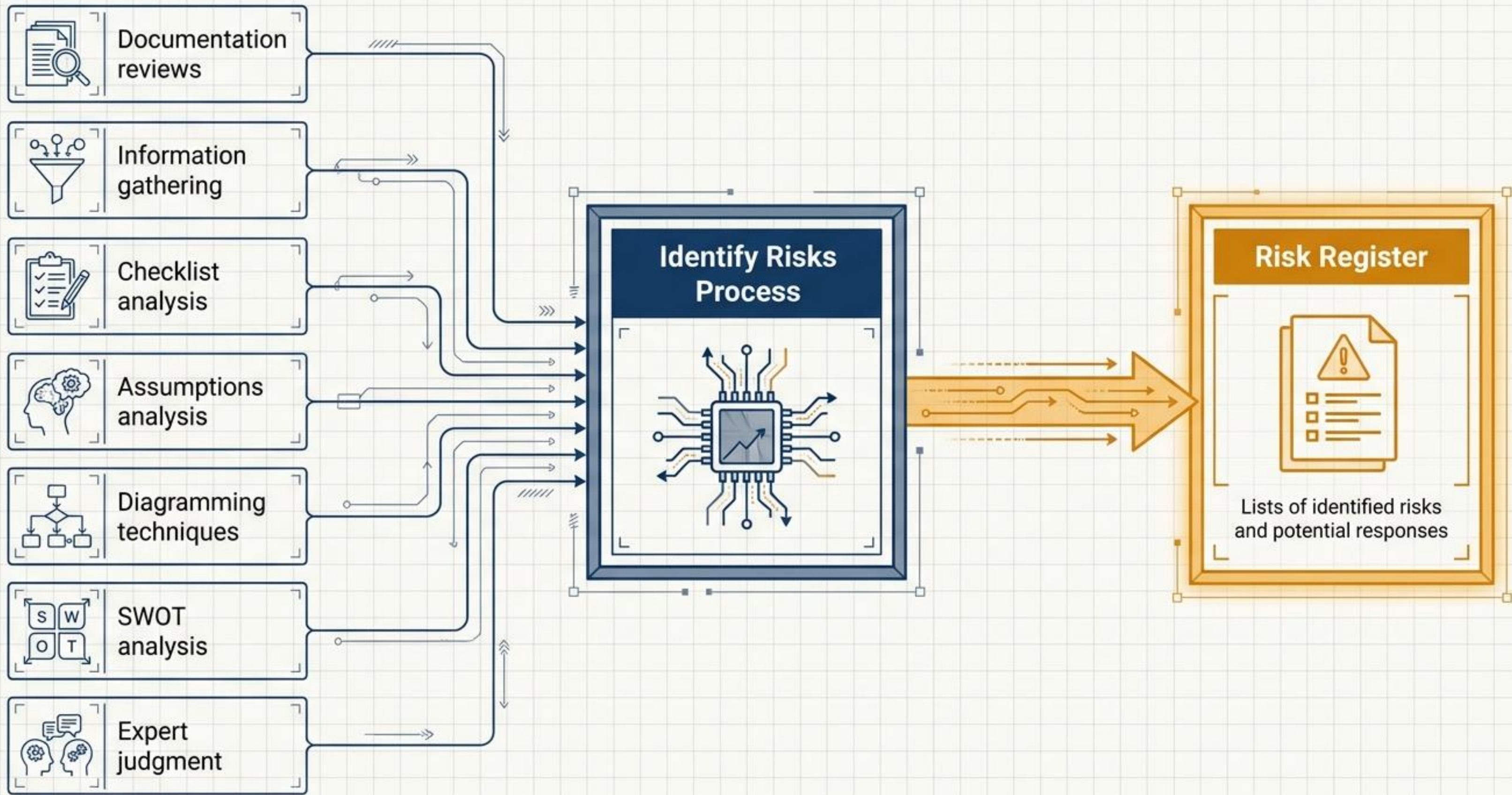


Certyfikat PMP: Risk Management Tools

A Diagnostic Playbook for
Identifying and Mapping
Project Risk

dr Zbigniew Galar





A



Documentation Reviews

Reviewing project plans, assumptions, procurement docs, and historical info.

PMP Key Insight

Look for gaps in quality, completeness, and consistency.

(e.g., An exceptionally documented schedule paired with a poorly documented budget signals a **high probability of risk**).

B



Assumptions Analysis

Validating assumptions made during project planning. False assumptions must be evaluated and scored as risks.

The 2-Factor Test

1

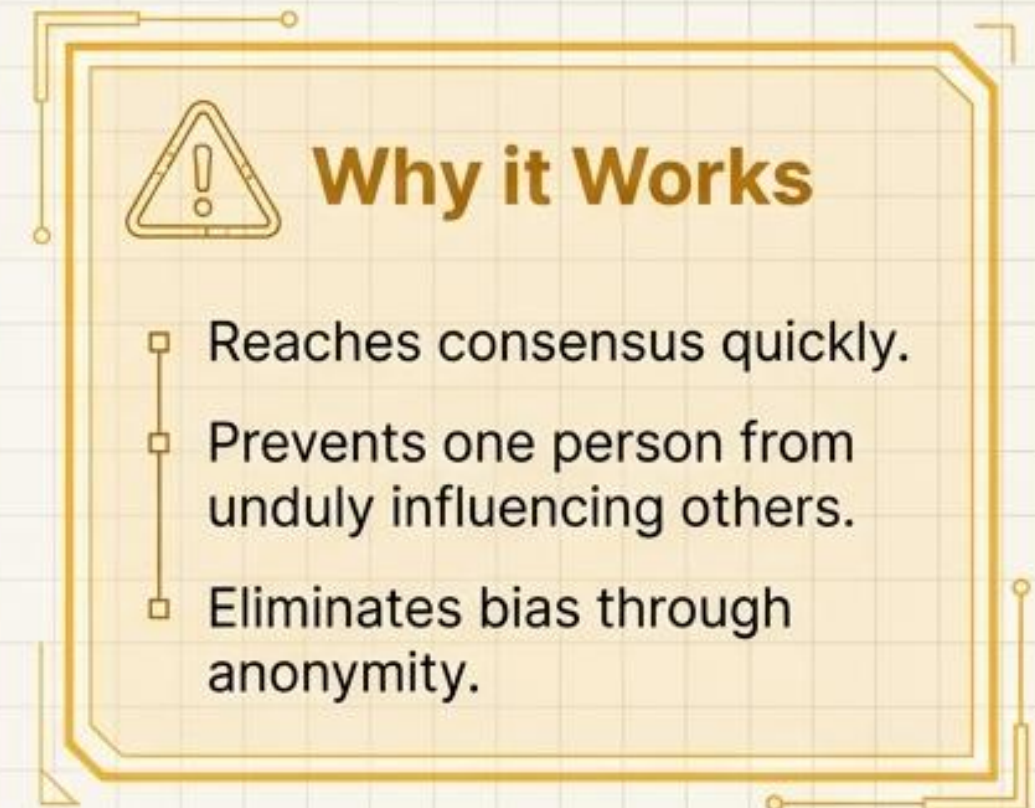
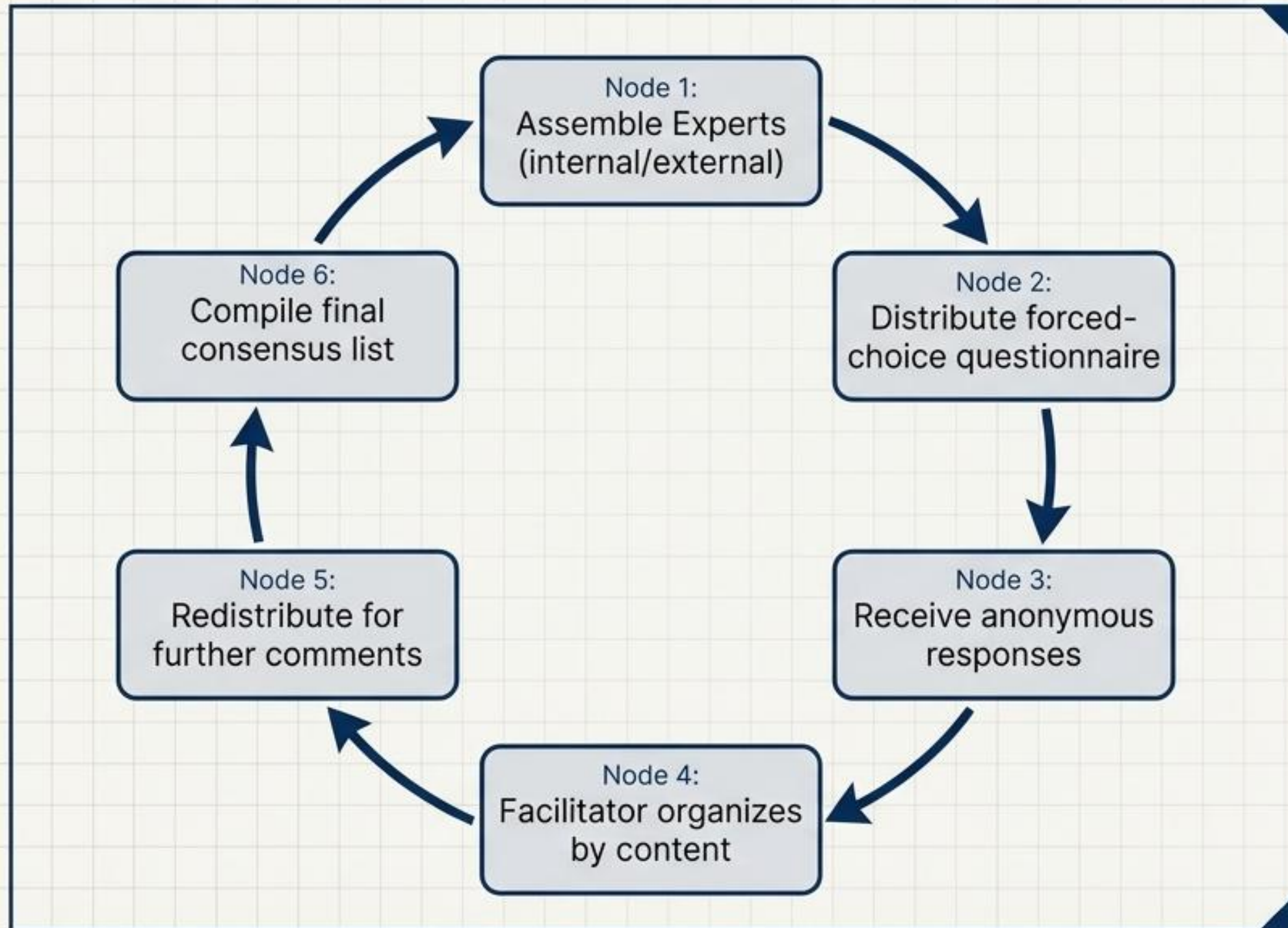
The **strength/validity** of the assumption.

2

The **consequences** to the project if the assumption is false.

Technique	Primary Execution Method	Key Advantage / Identifying Feature
Brainstorming	Face-to-face group session with SMEs and stakeholders.	One idea spawns another. Can use De Bono's Six Thinking Hats or lateral thinking.
Nominal Group Technique	Mass interview/group session.	Participants write risks independently, then the group ranks and prioritizes them in writing.
Interviewing	Direct Q&A sessions.	Leverages past experience of specialized experts. Uses WBS and assumptions list as prompts.
Root Cause Analysis	Deep-dive diagnostic.	Looks past the symptoms to find the underlying cause, defining the risk clearly for future response planning.

PMP Spotlight: The Delphi Technique

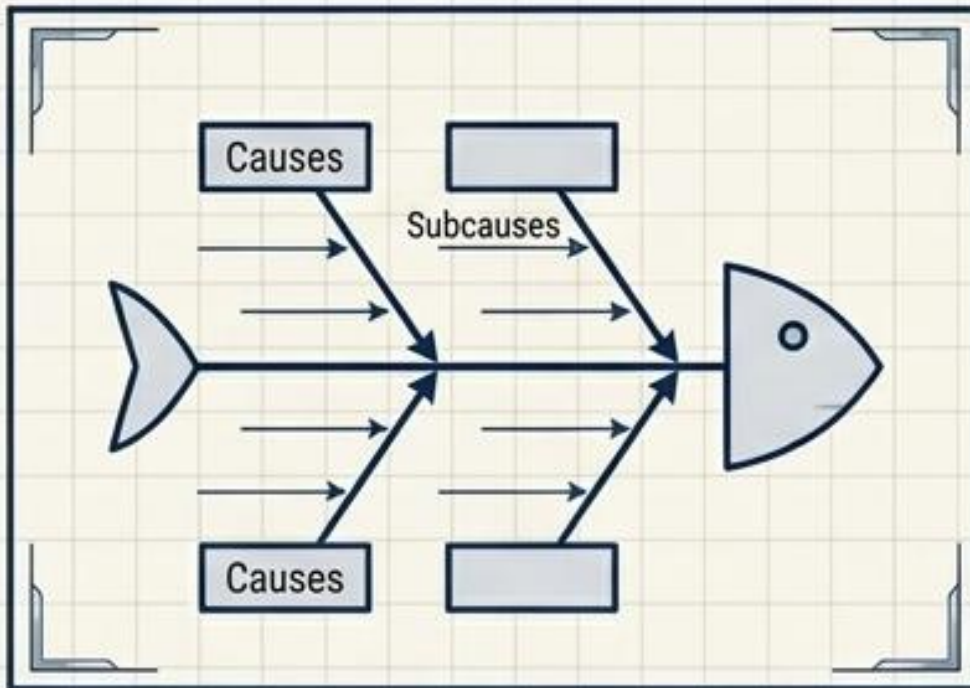


Internal vs. External Risk: SWOT Analysis

Origin (Internal vs. External)		
Impact (Positive vs. Negative)	Strengths	Opportunities
	What the organization does well. Associated with positive risks (opportunities to exploit).	Favorable external conditions.
	Weaknesses	Threats
	Areas for improvement. Associated with negative risks (threats to mitigate).	Harmful external conditions.

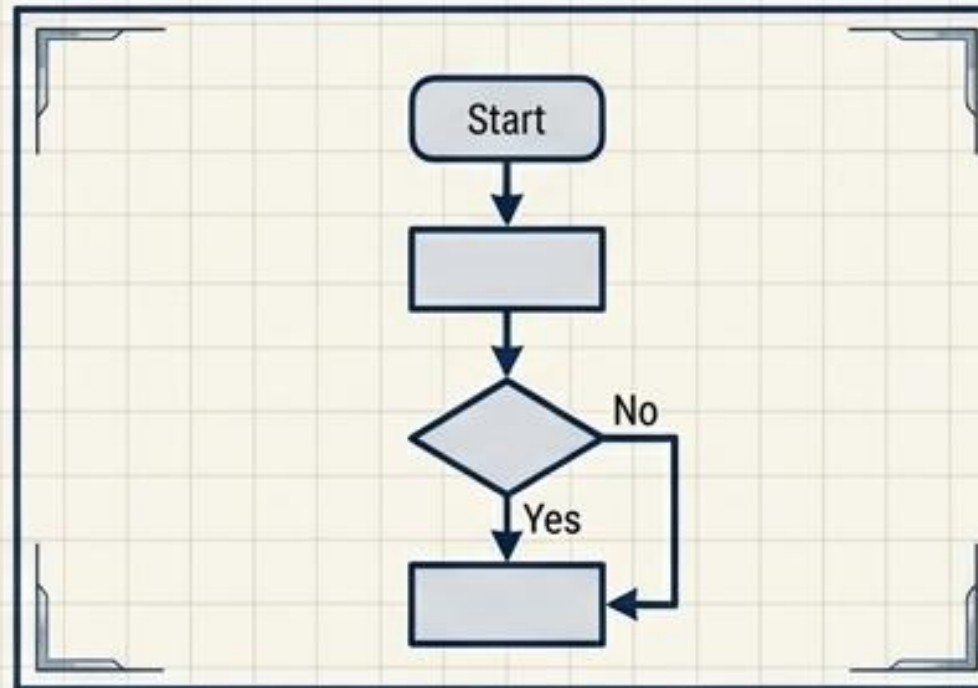
Note: SWOT (Internal-external analysis) is often combined with brainstorming to discover and document potential risks.

Visualizing Risk: 3 Diagramming Techniques



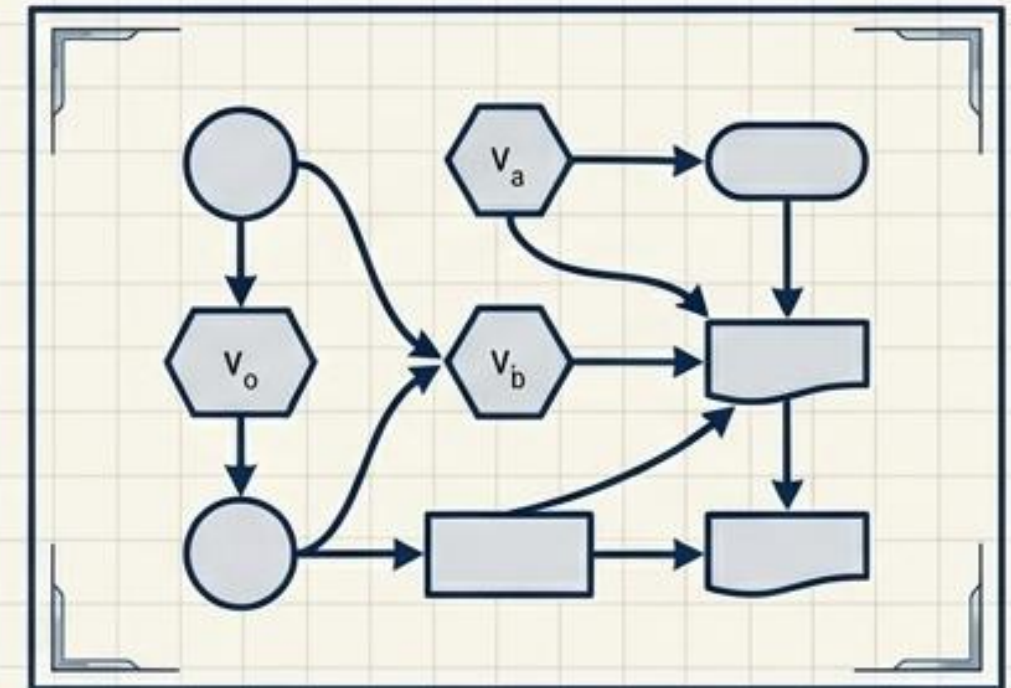
Cause-and-Effect Diagrams

(Also known as Fishbone or Ishikawa).
Shows the relationship between the effects of problems and their causes/subcauses.



System or Process Flowcharts

Shows logical steps to accomplish an objective, element relationships, and decision branches.

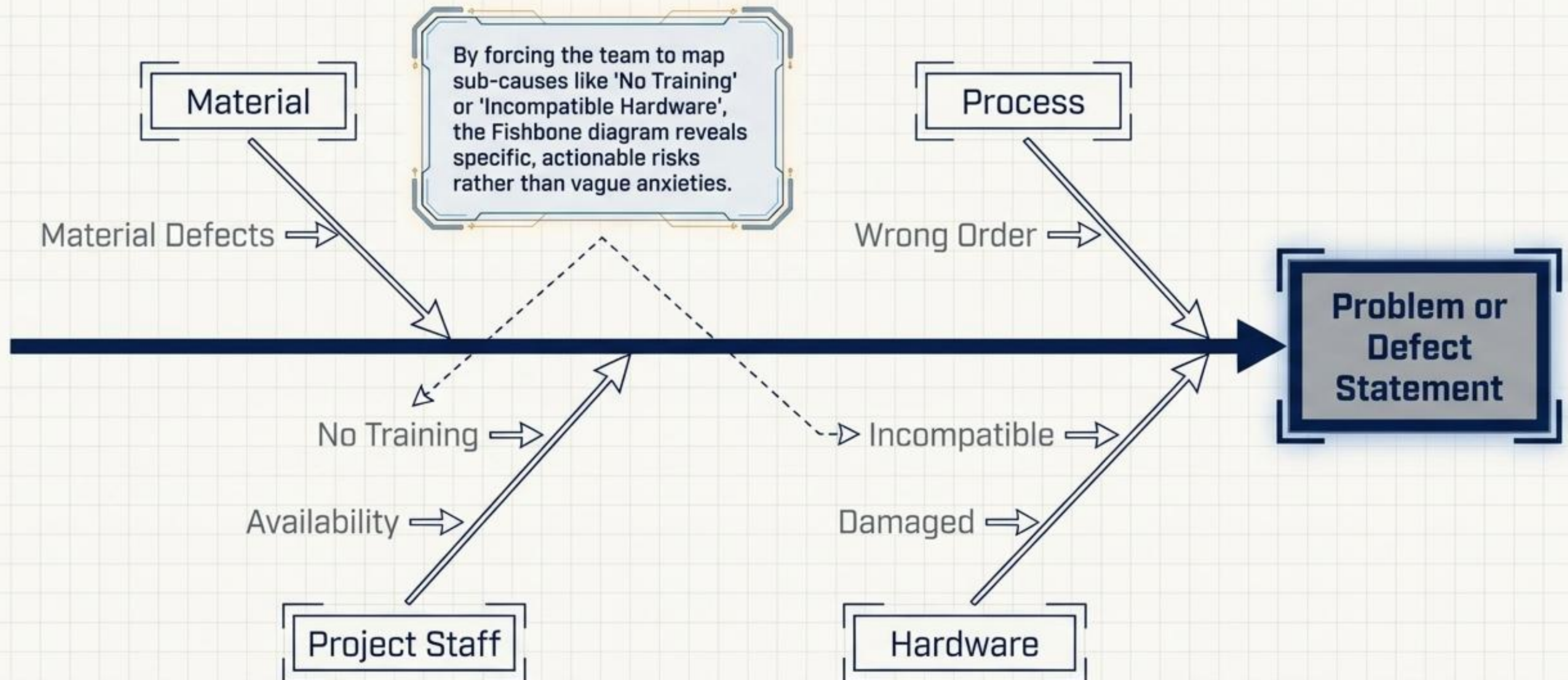


Influence Diagrams

Shows causal influences, time ordering, and relationships among project variables and outcomes.

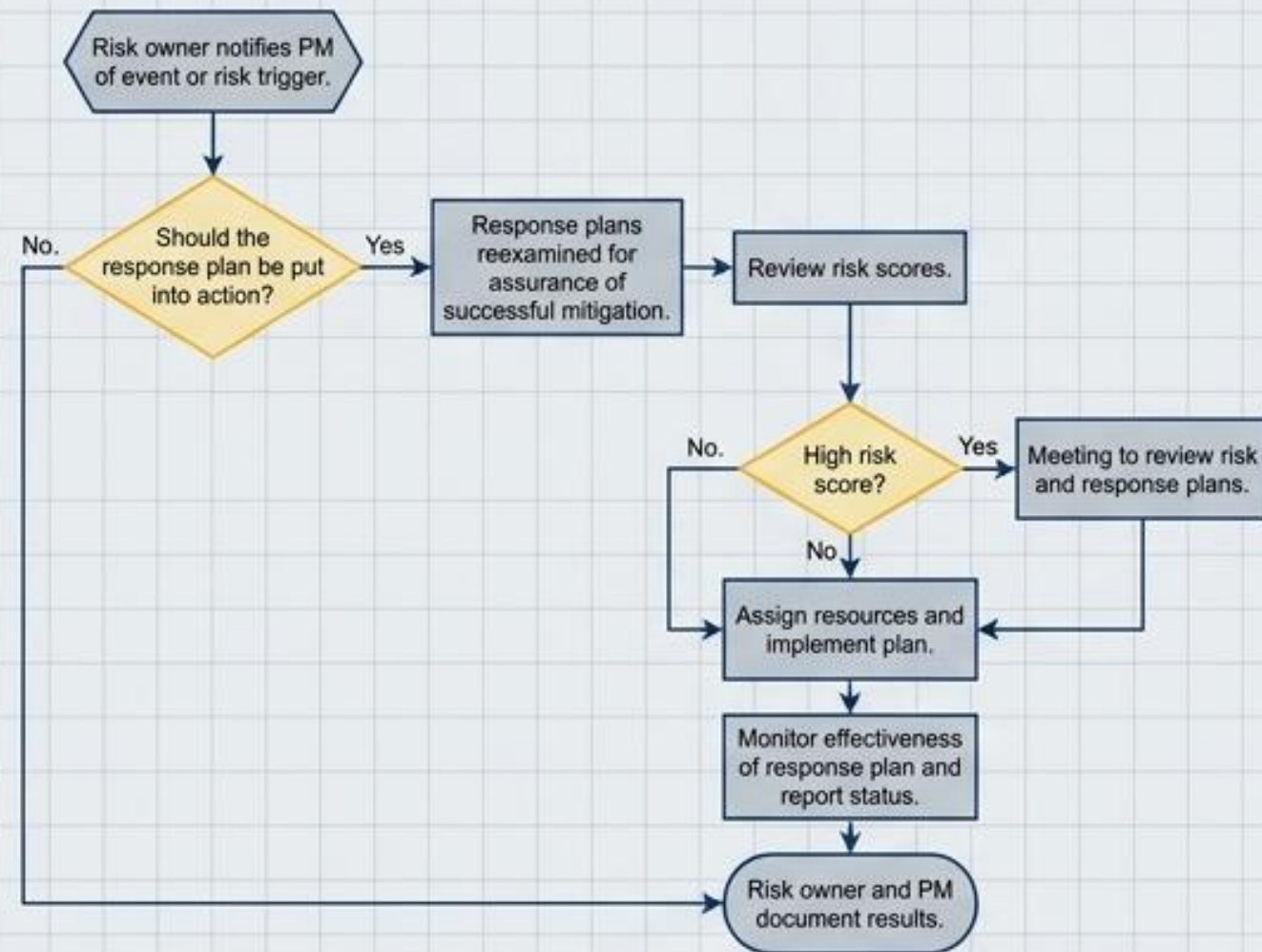
PMP Note: Cause-and-effect diagrams and flowcharts are used in both the Identify Risks process and the Perform Quality Control process.

Root Causes: The Ishikawa (Fishbone) Diagram



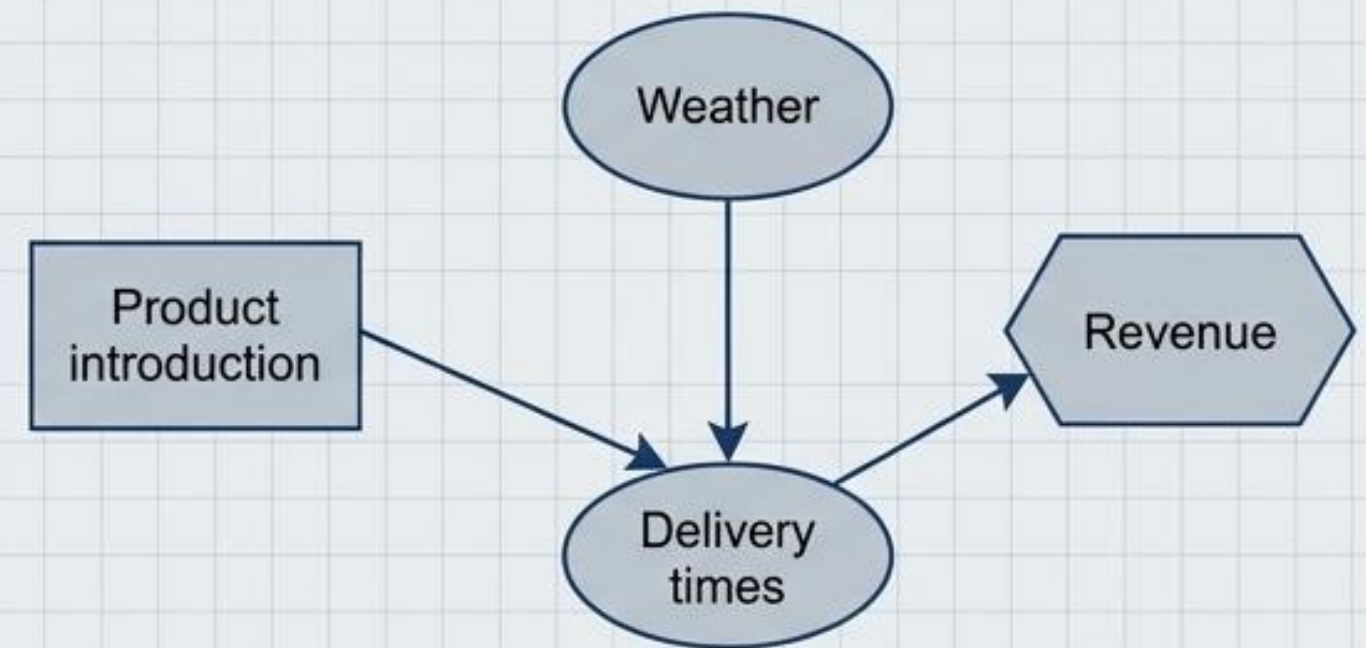
Logic & Variables: Flowcharts & Influence Diagrams

Process Flowcharts



Uses rectangles and parallelograms to step through a logical sequence. Look for risks hiding inside the 'Yes' and 'No' decision branches or logic loops.

Influence Diagrams

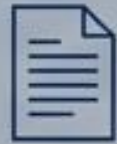


Visually depicts uncertainties and how variables influence each other over time. Example: Weather impacts Delivery times, which in turn impacts Revenue.

Experiential Tools: Checklists & Expert Judgment



Checklist Analysis



Concept: Built from historical information and the lowest level of the Risk Breakdown Structure (RBS).



The Catch: Fast and easy, but rarely exhaustive.



Rule: Never rely solely on checklists. Always update them at the end of the project with newly discovered risks.



Expert Judgment



Concept: Leverages individuals with experience in similar projects, specific business areas, or industry expertise.



The Catch: Human expertise carries blind spots.



Rule: The project manager must actively watch for and manage any bias the experts may have regarding the project or potential risk events.

The Output: Anatomy of a Risk Register

List of Identified Risks

Potential events and subsequent consequences. Assign tracking numbers.

List of Potential Responses

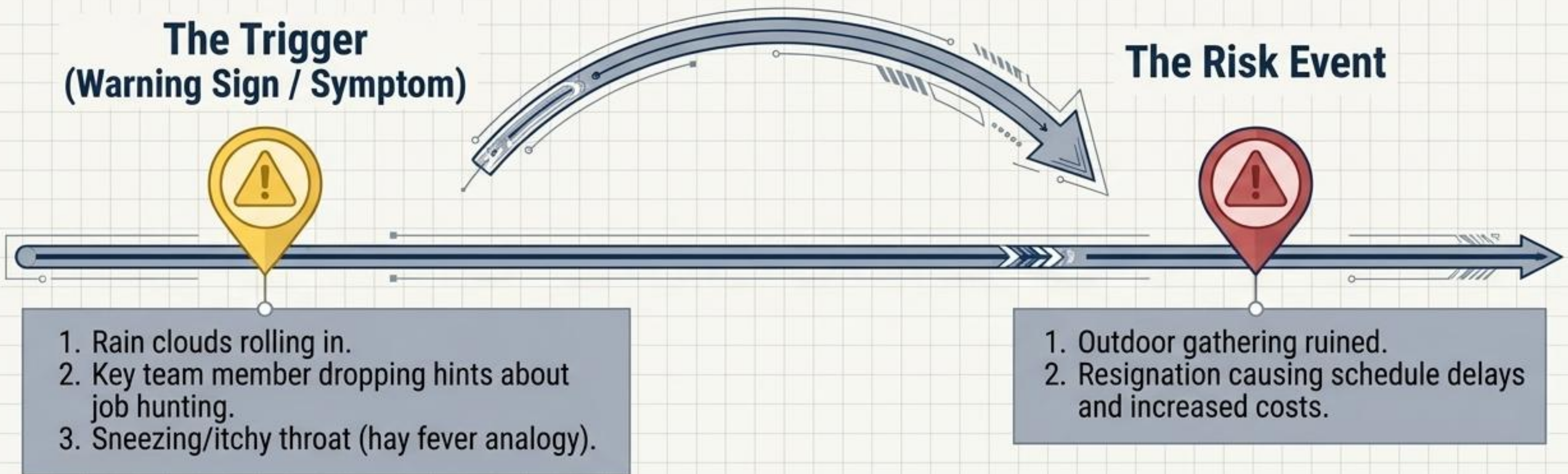
Sometimes identifying a risk immediately reveals the response. Document it now for use in the Plan Risk Responses process.

ID	Risk	Trigger	Event	Cause	Impact	Owner	Response plan
1	Infrastructure team is not available when needed	Predecessor tasks not completed on time	Operating system upgrade delayed	Equipment was not delivered on time	Schedule delay	Brown	Compress the schedule by beginning tasks in the next milestone while working on operating system upgrade

The Living Artifact

The register must be updated iteratively with new information as the project progresses.

Triggers: The Early Warning System



Key Takeaway: Triggers signal that a risk event is about to occur. Practically, they should be listed in the Risk Register immediately to serve as an early warning system.

Summary: The Risk Architect's Golden Rules



1. Identify Early:

The better the risk identification during the Planning stage, the more comprehensive the response plan will be. Do not skip this.



2. Mix Your Tools:

Never rely solely on historical checklists; combine them with dynamic group techniques like Brainstorming or Delphi.



3. Map the Roots:

Use Ishikawa diagrams and Root Cause Analysis to treat the underlying problem, not just the symptoms.



4. Watch for Bias:

Actively manage undue influence in group settings (use Delphi) and check Expert Judgment for personal bias.



5. Iterate the Register:

The Risk Register is a living document. Update it continuously as new risks are identified and triggers activate.